

Modern Quantum Field Theory

Paul Haskins
22338905

haskinsp@tcd.ie

Contents

1	Introduction	2
1.1	Quantum Mechanics via the Path Integral	2
1.2	Single Particle in 1D	3
1.3	Transition Amplitude for Finite Time	4
1.4	Correlation/Green's Functions	5
1.5	Functional Derivatives	7
1.5.1	Vacuum-Vacuum Expectation Value	8
1.6	Quantum Mechanics for Many Particles	9

The following is some lecture notes on Feynman's Path Integral Formalism of Quantum Field Theory, based off the lecture notes of Dr. Matthias Dennemann.

Recommended Reading:

- Peskin and Schroder
- Sredncki
- Field Theory, A modern Primer Ramond

1 Introduction

For these lecture notes we consider the functional formalism/functional quantisation of quantum field theory [1]. This formalism treats the Lagrangian as the fundamental object and then one can Legendre transform to the Hamiltonian; as opposed to the canonical approach which treats the Hamiltonian as the fundamental object.

1.1 Quantum Mechanics via the Path Integral

Consider a single non-relativistic particle that is defined in a $(0, 1)$ dimensional space. Let q, p be the position and momentum; $\hat{q}, \hat{p}$ are the position operator and momentum operators; $|q\rangle, |p\rangle$ are the eigenstates of the operators. Thus we have

$$\hat{q} |q\rangle = q |q\rangle \quad \hat{p} |p\rangle = p |p\rangle \quad (1.1)$$

All objects introduced above are defined on the Hilbert space, in general, $\langle \phi | \psi \rangle \in \mathbb{C}$. For all states in the Hilbert space there exists a linear decomposition in terms of $|p\rangle$ and $|q\rangle$.

Definition 1.1.1 Fourier Expansion of Momentum and Position States.

$$\begin{aligned} |\psi\rangle &:= \int dq |q\rangle \langle q | \psi \rangle & \langle q | \psi \rangle &:= \psi(q) \\ |\psi\rangle &:= \int dp |p\rangle \langle p | \psi \rangle & \langle p | \psi \rangle &:= \tilde{\psi}(p). \end{aligned}$$

These states obey a completeness relation.

Definition 1.1.2 Completeness Relation.

$$\int dq |q\rangle \langle q| = \int dp |p\rangle \langle p| = \mathbf{1} \quad (1.2)$$

Remark 1.1.3 Inner Product of States.

We can express the inner product in terms of the Fourier transform

$$\langle q | \psi \rangle = \psi(q) = \int \frac{dp}{\sqrt{2\pi\hbar}} e^{i\frac{pq}{\hbar}} \tilde{\psi}(p) \quad (1.3)$$

$$\langle q | p \rangle = \psi(p) = \int \frac{dp'}{\sqrt{2\pi\hbar}} e^{i\frac{p'q}{\hbar}} \langle p' | p \rangle \quad \langle p' | p \rangle = \delta(p - p'). \quad (1.4)$$

Theorem 1.1.4 Momentum Position Inner Product.

$$\langle q | p \rangle = \frac{e^{i\frac{pq}{\hbar}}}{\sqrt{2\pi\hbar}} \quad (1.5)$$

Proof.

XXX

□

1.2 Single Particle in 1D

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q}) \quad \hat{p} = -i\hbar \frac{\partial}{\partial q} \quad (1.6)$$

$$\left\langle q \left| \hat{p} \right| \psi \right\rangle = -i\hbar \frac{\partial}{\partial q} \psi(q) \quad (1.7)$$

In general we have

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \quad (1.8)$$

If the state is time independent then

$$|\psi(t)\rangle = e^{-i\hat{H}(t-t_0)} |\psi(t_0)\rangle \quad (1.9)$$

Define

$$\psi(q, t) := \left\langle q \left| \psi(t) \right\rangle \quad (1.10)$$

If we have an initial state $|q_a\rangle$ at time t_a and a final state $|q_b\rangle$ at time t_b ,

$$\begin{aligned} \psi(q_b, t_b) &= \left\langle q_b \left| \psi(t_b) \right\rangle = \left\langle q_b \left| e^{-i\hat{H}\left(\frac{t_b-t_a}{\hbar}\right)} \right| \psi(t_a) \right\rangle \\ &= \int dq_a \langle q_b | e^{-i\hat{H}(t_b-t_a)/\hbar} \left\langle q_a \left| \psi(t_a) \right\rangle \right\rangle. \end{aligned}$$

We define the propagator amplitude

$$K(q_a, q_b, t_a, t_b) = \left\langle q_b \left| e^{-i\hat{H}(t_b-t_a)/\hbar} \right| q_a \right\rangle \quad (1.11)$$

If we consider the time difference between the states to be small $t_b - t_a = \varepsilon$ and omitt terms $\mathcal{O}(\varepsilon^2)$.

$$\begin{aligned} K(q_a, q_b, \varepsilon) &= \left\langle q_b \left| e^{-i\hat{H}\varepsilon/\hbar} \right| q_a \right\rangle \\ &= \left\langle q_b \left| e^{-\frac{i\varepsilon}{\hbar} \left[\frac{\hat{p}^2}{2m} + V(\hat{q}) \right]} \right| q_a \right\rangle. \end{aligned}$$

We can use the BCH formula

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A} + \hat{B} + \frac{1}{2!} [\hat{A}, \hat{B}] + \dots} \quad (1.12)$$

In our case, we have $\hat{A} = -i\frac{\varepsilon}{\hbar} \frac{\hat{p}^2}{2m}$, $\hat{B} = -i\frac{\varepsilon}{\hbar} V(\hat{q})$ and the commutator term is of $\mathcal{O}(\varepsilon^2)$

$$\begin{aligned} K &= \left\langle q_b \left| e^{-i\frac{\varepsilon}{\hbar} \frac{\hat{p}^2}{2m}} e^{-i\frac{\varepsilon}{\hbar} V(\hat{q})} \right| q_a \right\rangle + \mathcal{O}(\varepsilon^2) \\ &= e^{-i\frac{\varepsilon}{\hbar} V(q_a)} \int dp \left\langle q_a \left| e^{-\frac{i\varepsilon}{\hbar} \frac{\hat{p}^2}{2m}} \right| p \right\rangle \left\langle p \left| q_a \right\rangle \right\rangle \\ &= e^{-i\frac{\varepsilon}{\hbar} V(q_a)} \int dp e^{-\frac{i\varepsilon}{\hbar} \frac{p^2}{2m}} \left\langle q_a \left| p \right\rangle \left\langle p \left| q_a \right\rangle \right\rangle \\ &= e^{-i\frac{\varepsilon}{\hbar} V(q_a)} \int \frac{dp}{2\pi\hbar} e^{-i\frac{\varepsilon}{\hbar} \frac{p^2}{2m} + \frac{i\varepsilon}{\hbar} p(q_b - q_a)}. \end{aligned}$$

where we used the completeness relation (1.2) in line 2 and using the definitions of the inner products (1.3) in the final line.

Now we may use the Gaussian type integrals

$$\int_{-\infty}^{\infty} e^{-ax^2} = \sqrt{\frac{\pi}{a}} \quad (1.13)$$

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}. \quad (1.14)$$

In our case we have $a = \frac{i\varepsilon}{\hbar 2m}$ and $b = \frac{i}{\hbar} (q_b - q_a)$. Thus, we have

$$K = N e^{i\frac{\varepsilon}{\hbar} \mathcal{L}(q_a)} \quad (1.15)$$

where we introduce

$$N := \sqrt{\frac{m}{2\pi i \hbar \varepsilon}}$$

$$\mathcal{L}(q_a) = \frac{i\varepsilon}{\hbar} \left[\frac{1}{2} m \left(\frac{q_b - q_a}{\varepsilon} \right)^2 - V(q_b) \right].$$

1.3 Transition Amplitude for Finite Time

Let $t_b - t_a = n\varepsilon$. For $\varepsilon \rightarrow 0$, $m \rightarrow \infty$, $t_b - t_a$ is finite. Take the position of an arbitrary intermediate state q_J , $J = 1, \dots, n-1$. Let $q_0 = q_a$, $q_b = q_n$ and sum over all possible intermediate states¹

$$\begin{aligned} K(q_b, q_a, t_b - t_a) &= \int \prod_{J=1}^{n-1} dq_J \left\langle q_b \left| e^{-\frac{i}{\hbar} \varepsilon \hat{H}} \right| q_{n-1} \right\rangle \cdot \left\langle q_{n-1} \left| e^{-\frac{i}{\hbar} \varepsilon \hat{H}} \right| q_{n-2} \right\rangle \cdot \dots \cdot \left\langle q_1 \left| e^{-\frac{i}{\hbar} \varepsilon \hat{H}} \right| q_a \right\rangle \\ &= \int \prod_{J=1}^{n-1} dq_J K(q_b, q_{n-1}, \varepsilon), \dots, K(q_1, q_a, \varepsilon) \\ &= \int \left[\prod_{J=1}^{n-1} dq_J \right] N^n e^{i\frac{\varepsilon}{\hbar} \sum_{J=1}^{n-1} \mathcal{L}(q_J)}. \end{aligned}$$

In the limit $\varepsilon \rightarrow 0$, $n \rightarrow \infty$

$$K(q_b, q_a, t_b - t_a) = n \int \mathcal{D}q e^{\frac{i}{\hbar} S[q]} \quad (1.16)$$

where

$$S[q] = \int_{t_a}^{t_b} dt \mathcal{L}[\dot{q}] \quad (1.17)$$

From now on we set $\hbar \rightarrow 1$.

¹One can consider the square of the amplitude and interpret this as the probability for each given transition from one intermediate state to the next intermediate state.

1.4 Correlation/Green's Functions

Definition 1.4.1 One Point Functions.

One point functions are given by $\langle q_b, t_b | \hat{q}(t_*) | q_a, t_a \rangle$ for some intermediate time t_* . $\hat{\phi}(t_*) | q \rangle = q(t_*) | q \rangle$.

$$\begin{aligned} \langle q_b, t_b | \hat{q}(t_*) | q_a, t_a \rangle &= \int \prod_{J=1}^{n-1} dq_J \langle q_b | e^{-\frac{i}{\hbar} \varepsilon \hat{H}} | q_{n-1} \rangle \cdot \dots \cdot \langle q_m | e^{-\frac{i}{\hbar} \varepsilon \hat{H}} | q(t_*) \rangle \cdot \langle q_* | e^{-\frac{i}{\hbar} \varepsilon \hat{H}} | q_{m+1} \rangle \\ &\quad \cdot \dots \cdot \langle q_1 | e^{-\frac{i}{\hbar} \varepsilon \hat{H}} | q_a \rangle \\ &= \int \prod_{J=1}^{n-1} dq_J q(t_*) e^{i\varepsilon \sum_{J=1}^{n-1} \mathcal{L}(p)}. \end{aligned}$$

Definition 1.4.2 Two Point Functions.

The two point function is given by

$$\langle q_a t_a | \hat{q}(t_1) \hat{q}(t_2) | q_a t_a \rangle = n \int \mathcal{D}q q(t_1) q(t_2) e^{iS[q]} \quad (1.18)$$

Here we have two operators at different times, thus we need to consider the time ordering. We demand $t_a t_1 < t_2 < t_a$. Define the time ordering operator

$$\hat{T} \{ \hat{q}(t_1), \hat{q}(t_2) \} = \begin{cases} \hat{q}(t_1) \hat{q}(t_2) & \text{if } t_1 < t_2 \\ \hat{q}(t_2) \hat{q}(t_1) & \text{if } t_2 < t_1 \end{cases} \quad (1.19)$$

XXX what if $t_1 = t_2$?

Definition 1.4.3 n-Point Functions.

$$\langle q_b t_b | \hat{T} \{ \hat{q}(t_1), \dots, \hat{q}(t_m) \} | q_a t_a \rangle = \mathcal{N} \int \mathcal{D}q q(t_1) \dots q(t_m) e^{iS[q]} \quad (1.20)$$

Remark 1.4.4 Ground State and Vacuum State.

We distinguish between the ground state $|\Omega\rangle$ and the vacuum state $|0\rangle$, in general $|\Omega\rangle \neq |0\rangle$. The special case of $|\Omega\rangle = |0\rangle$ is that of the case when the lowest energy state of the qft contains no particle excitations e.g. hypothetical empty space and QED. When a system possesses a symmetry, but the lowest energy configuration requires choosing a specific orientation or value, you get infinitely many equivalent ground states. This leads directly to spontaneous symmetry breaking.

$$\langle \Omega | \hat{T} \{ \hat{q}(t_1), \dots, \hat{q}(t_m) \} | \Omega \rangle \quad (1.21)$$

If we consider time interval $[-T, T]$ and take $T \rightarrow \infty$. This correlation function is the vacuum-to-vacuum n -point function.

Given a Hamiltonian $\hat{H}$, the eigenvalues and eigenstates are

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle \quad n = 0, 1, \dots \quad (1.22)$$

such that $E_n \leq E_{n+1}$ and let E_0 be the lowest energy of the system, and the associated state is the ground state $|\psi_0\rangle = |\Omega\rangle$.

Since $\hat{H}$ is the Hamiltonian operator, the eigenstates form a complete basis for the system. A state can be written as

$$|q_a, t\rangle = e^{i\hat{H}t} |q_a\rangle \quad (1.23)$$

where the reference time is chose to be zero.

We input a complete set of states

$$\begin{aligned} |q_a, t\rangle &= \sum_n |\psi_n\rangle \langle \psi_n | q_a \rangle \\ &= \sum_n e^{iE_n t} |\psi_n\rangle \langle \psi_n | q_a \rangle \\ &= e^{iE_0 t} |\Omega\rangle \langle \Omega | q_a \rangle \\ &= e^{iE_0 t} |\Omega\rangle \langle \Omega | q_a \rangle + \sum_{n>0} e^{iE_n t} |\psi_n\rangle \langle \psi_n | q_a \rangle. \end{aligned}$$

We want to isolate the ground state contribution. We only want that term at $t = -T$ for $T \rightarrow \infty$. In order to get rid of the $n > 0$ terms, we can take the limit $T \rightarrow (1 - i\varepsilon)\infty$, i.e. slightly tilt along the imaginary direction.

$$e^{-iE_n(1-i\varepsilon)T} = e^{-iE_n T} e^{-\varepsilon E_n T} \quad (1.24)$$

so they still oscillate but they are dampened. Since $E_0 \leq E_n, \forall n$, in $T \rightarrow (1 - i\varepsilon)\infty$, the $n > 0$ terms are exponentially suppressed with respect to the ground contribution, with exponent given by $E_n - E_0$.

$$|q_a, -T\rangle \underset{T \rightarrow (1-i\varepsilon)\infty}{\sim} e^{iE_0 T} |\Omega\rangle \langle \Omega | q_a \rangle + \dots \quad (1.25)$$

We can thus write the ground state as

$$|\Omega\rangle \approx \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{iE_0 T} |q_a, -T\rangle}{\langle \Omega | q_a \rangle} \quad (1.26)$$

Its conjugate is given by

$$\langle \Omega | = \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{it_0 T} \langle q_b, T |}{\langle q_b | \Omega \rangle} \quad (1.27)$$

Definition 1.4.5 Ground state Vacuum Expectation Value.

We demand $\langle \Omega | \Omega \rangle = 1$

Theorem 1.4.6 Ground State Normalisation Coefficient.

$$\mathcal{N} = \left[\lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T} \int \mathcal{D}q e^{iS[q]}}{\langle \Omega | q_a \rangle \langle \Omega | q_b \rangle} \right]^{-1} \quad (1.28)$$

Proof.

Given Def. 1.4.5 and we notice if $n = 0$ in (1.28) we have

$$1 = \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{\mathcal{N} e^{2iE_0 T \int \mathcal{D}q e^{iS[q]}}}{\langle \Omega | q_a \rangle \langle \Omega | q_b \rangle}$$

$$\Rightarrow \mathcal{N} = \left[\lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T \int \mathcal{D}q e^{iS[q]}}}{\langle \Omega | q_a \rangle \langle \Omega | q_b \rangle} \right]^{-1}.$$

□

Theorem 1.4.7 Ground State Expectation Value.

$$\langle \Omega | \hat{T} \{ \hat{q}(t_1), \dots, \hat{q}(t_n) \} | \Omega \rangle = \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T \int \mathcal{D}q \prod_{j=1}^n q(t_j)} e^{iS[q]}}{\int \mathcal{D}q e^{iS[q]}}. \quad (1.29)$$

Proof. We have the ground state expectation value given by

$$\begin{aligned} \langle \Omega | \hat{T} \{ \hat{q}(t_1), \dots, \hat{q}(t_n) \} | \Omega \rangle &= \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T} \langle q_b, T | \hat{T} \{ \hat{q}(t_1), \dots, \hat{q}(t_n) \} | q_n, -T \rangle}{\langle \Omega | q_a \rangle \langle \Omega | q_b \rangle} \\ &= \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T} \mathcal{N} \int \mathcal{D}q q(t_1) \dots q(t_n) e^{iS[q]}}{\langle \Omega | q_a \rangle \langle \Omega | q_b \rangle} \\ &= \lim_{T \rightarrow (1-i\varepsilon)\infty} \frac{e^{2iE_0 T} \int \mathcal{D}q q(t_1) \dots q(t_n) e^{iS[q]}}{\int \mathcal{D}q e^{iS[q]}}. \end{aligned}$$

where we used (1.28) and found the result which matches (1.29). □

1.5 Functional Derivatives

Definition 1.5.1 Functional Derivatives.

$$\frac{\delta f(t_2)}{\delta f(t_1)} := \delta(t_2 - t_1) \quad (1.30)$$

Remark 1.5.2 Continuous Analogue.

This is the Continuous analogue of

$$\frac{\partial x^i}{\partial x^j} = \delta_j^i \quad (1.31)$$

1.5.1 Vacuum-Vacuum Expectation Value

Consider the transformation

$$S[q] \rightarrow S[q] + i \int_{-\infty}^{\infty} dt J(t) q(t) \quad (1.32)$$

where $J(t)$ is the source current.

Thus,

$$\int \mathcal{D}q \prod_{j=1}^n e^{iS[q]} = \left\{ \int \mathcal{D}q \frac{1}{i} \frac{\delta}{\delta J(t_1)} \times \dots \times \frac{1}{i} \frac{\delta}{\delta J(t_n)} e^{iS[q] + i \int_{-\infty}^{\infty} dt J(t) q(t)} \right\} \Big|_{J=0} \quad (1.33)$$

Example 1.5.3.

$$\begin{aligned} \frac{1}{i} \frac{\delta}{\delta J(t_1)} e^{i \int_{-\infty}^{\infty} J(t) q(t)} &= \frac{1}{i} e^{i \int_{-\infty}^{\infty} dt J(t) q(t)} \times \frac{\delta}{\delta J(t_1)} \left[i \int_{-\infty}^{\infty} dt J(t) q(t) \right] \\ &= e^{i \int_{-\infty}^{\infty} dt J(t) q(t)} \times \left[\int_{-\infty}^{\infty} dt \delta(t - t_1) q(t) \right] \\ &= e^{i \int_{-\infty}^{\infty} dt J(t) q(t)} q(t_1). \end{aligned}$$

Definition 1.5.4 Generating Functional.

$$Z[J] := \int \mathcal{D}q(t) e^{iS[q] + i \int_{-\infty}^{\infty} dt J(t) q(t)} \quad (1.34)$$

Thus, the generating functional evaluated at zero source is the partition function:²

$$Z[0] = \int \mathcal{D}q e^{iS[q]} \quad (1.35)$$

Remark 1.5.5.

One can write

$$\int \mathcal{D}q q(t_1) \dots q(t_n) e^{iS[q]} = \left(\frac{1}{i} \frac{\delta}{\delta J(t_1)} \dots \frac{1}{i} \frac{\delta}{\delta J(t_n)} Z[J] \right) \Big|_{J=0} \quad (1.36)$$

The vacuum-vacuum correlation function of a time ordered product of hermitian operators is³

$$\left\langle \Omega \left| \hat{T} \{ \hat{q}(t_1) \} \dots \hat{q}(t_n) \right| \Omega \right\rangle = \left[\frac{1}{i^n} \frac{\delta}{\delta J(t_1)} \dots \frac{\delta}{\delta J(t_n)} \frac{Z[J]}{Z[0]} \right] \Big|_{J=0} \quad (1.37)$$

²Bekeinstein entropy?

³why hermitian?

1.6 Quantum Mechanics for Many Particles

Bibliography

- [1] R. P. Feynman. “Space-Time Approach to Non-Relativistic Quantum Mechanics”. In: *Reviews of Modern Physics* 20.2 (1948), pp. 367–387. DOI: [10.1103/RevModPhys.20.367](https://doi.org/10.1103/RevModPhys.20.367).